TransitOps Test Plan

Test Plan & Test Cases

TransitOps — Fleet & Logistics Operations Platform

Version: 1.0 **Companion documents:** SRS, TRD, Backend Schema, API Documentation

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1. Test Plan

1.1 Objective

Verify that TransitOps enforces its documented business rules (BR-1 through BR-9), RBAC permissions, and API contracts correctly, prior to release.

1.2 Scope

In scope: Auth, Vehicles, Drivers, Trips (full lifecycle), Fuel/Expenses, Maintenance, Reports, Notifications, RBAC enforcement across all endpoints. **Out of scope:** AI stub endpoints (`/ai/*` — not real logic), load/performance testing, third-party SMTP delivery verification.

1.3 Current Automation State

No automated test suite exists in the repository at time of writing (CI runs `npm test --if-present`, currently a no-op). This document defines the test cases that should be automated (e.g. Jest + Supertest against an in-memory MongoDB) and can also be executed manually via Postman/curl/Swagger UI in the interim.

1.4 Test Levels

Level	Approach
Unit	`tripService` methods (createTrip, dispatchTrip, completeTrip, cancelTrip, updateTripStatus, scheduling-conflict check) tested in isolation with mocked repository
Integration	Controller → Service → Repository → MongoDB, via Supertest against a running Express app + in-memory Mongo
API / E2E	Full request/response cycles including auth, RBAC, and validation, against a seeded test database
Manual/Exploratory	UI walkthrough per role, using seeded demo accounts

1.5 Test Environment

- Node 18/20, MongoDB 6.0 (test instance, separate from dev data)
- Seeded via `npm run seed` or a dedicated test fixture set
- `.env.test` with a disposable `JWT_SECRET` and `MONGO_URI`

1.6 Entry / Exit Criteria

Entry: Backend builds and starts; test database reachable; seed data or fixtures loaded. **Exit:** All P1 (business-rule and RBAC) test cases pass; no open Sev-1/Sev-2 defects.

1.7 Severity Definitions

Severity	Definition
Sev-1	Business rule not enforced (e.g. overloaded vehicle dispatched) or data corruption

Sev-2	RBAC bypass, incorrect status code/response shape
Sev-3	Cosmetic/UX issue, non-blocking

2. RBAC Test Matrix

Legend: ✅ allowed · ❌ 403 expected · — endpoint doesn't apply

Endpoint (write ops)	fleet_manager	dispatcher	safety_officer	financial_analyst	viewer
POST/PUT/DELETE `/vehicles`	✅	❌	❌	❌	❌
POST/PUT/DELETE `/drivers`	❌	❌	✅	❌	❌
POST `/trips`, dispatch/complete/cancel/status	✅	✅	❌	❌	❌
POST `/maintenance`, close	✅	❌	✅	❌	❌
POST `/fuel-expenses/fuel`, `/expenses`	✅	❌	❌	✅	❌
GET `/reports/driver-performance`, `/reports/export/trips`	✅	❌	❌	✅	❌
GET on any list/detail endpoint	✅	✅	✅	✅	✅

Every write endpoint above also requires **TC-RBAC-00: request without a JWT → 401**, tested once generically and assumed for all subsequent cases.

3. Detailed Test Cases

3.1 Authentication

ID	Title	Steps	Expected Result
TC-AUTH-01	Register with valid data	POST `/auth/register` with valid name/email/password/phone	201, token + user returned
TC-AUTH-02	Register with duplicate email	Register twice with same email	400, "Email already registered"
TC-AUTH-03	Register with short password	password = "abc123" (6 chars)	400, Zod validation error on `password`
TC-AUTH-04	Login with correct credentials	POST `/auth/login`	200, token returned
TC-AUTH-05	Login with wrong password	Correct email, wrong password	401, "Invalid email or password"
TC-AUTH-06	Access protected route without token	GET `/auth/me` no header	401
TC-AUTH-07	Access protected route with expired token	Use token past `JWT_EXPIRE`	401, "Token expired"
TC-AUTH-08	Refresh an expired token	POST `/auth/refresh` with expired-but-valid-signature token	200, new token issued
TC-AUTH-09	Forgot password for unknown email	POST `/auth/forgot-password` with non-existent email	200 generic success message (no enumeration)
TC-AUTH-10	Reset password with reused token after password change	Use a reset token issued before a prior successful reset	400, invalid/expired token

3.2 Vehicles

ID	Title	Steps	Expected Result
TC-VEH-01	Create vehicle as fleet_manager	POST `/vehicles` valid body	201, vehicle with `status: available`
TC-VEH-02	Create vehicle as dispatcher	Same body, dispatcher token	403
TC-VEH-03	Duplicate registration number	Create two vehicles with same `registrationNumber`	400, duplicate key error (field named in message)
TC-VEH-04	Missing required field	Omit `maxLoadCapacity`	400, Zod validation error
TC-VEH-05	Get vehicle by invalid id	GET `/vehicles/000000000000000000000000`	404, “Resource not found”
TC-VEH-06	List caching	GET `/vehicles` twice within 300s	Second response served from cache (verify via timing/log, not user-visible difference)
TC-VEH-07	Cache invalidation on write	POST a vehicle, then GET list	New vehicle appears immediately (cache cleared)
TC-VEH-08	Soft delete	DELETE `/vehicles/:id` then GET `/vehicles`	Vehicle absent from list; document still exists in DB with `deleted: true`

3.3 Drivers

ID	Title	Steps	Expected Result
TC-DRV-01	Create driver as safety_officer	POST `/drivers` valid body	201
TC-DRV-02	Create driver as fleet_manager	Same body, fleet_manager token	403
TC-DRV-03	Duplicate license number	Two drivers, same `licenseNumber`	400, duplicate key error
TC-DRV-04	PII encrypted at rest	Inspect raw Mongo document for a created driver	`licenseNumber` / `contactNumber` stored as ciphertext, not plaintext
TC-DRV-05	PII decrypted on read	GET `/drivers/:id`	`licenseNumber` / `contactNumber` returned as original plaintext
TC-DRV-06	Status enum mismatch check	PUT `/drivers/:id` with `status: "off_duty"`	Currently passes Mongoose enum but should be checked against Zod’s `on_leave` — **expected to reveal the known discrepancy (Backend Schema §7-1)**; log as regression once fixed
TC-DRV-07	Safety score bounds	Create driver with `safetyScore: 150`	400, Zod validation error (max 100)

3.4 Trip Lifecycle (highest priority — core business rules)

ID	Title	Steps	Expected Result
TC-TRP-01	Create trip, happy path	Valid vehicle (available), valid driver (available, valid license), cargo ≤ capacity, no schedule conflict	201, `status: draft`
TC-TRP-02	BR-1: retired vehicle	Create trip against a `retired` vehicle	400, “BR-1: Vehicle not found or is retired”
TC-TRP-03	BR-1: nonexistent vehicle	Create trip with a random valid-	400, BR-1 message

		format ObjectId for `vehicle`	
TC-TRP-04	BR-4: suspended driver	Create trip with a `suspended` driver	400, BR-4 message
TC-TRP-05	BR-4: expired license	Create trip with driver whose `licenseExpiryDate` is in the past	400, BR-4 message
TC-TRP-06	BR-5: overloaded cargo	`cargoWeight` > vehicle's `maxLoadCapacity`	400, BR-5 message
TC-TRP-07	Scheduling conflict — same vehicle	Create two trips for the same vehicle with overlapping `scheduledDate` – `eta` windows	Second create fails, 400, conflicting `tripCode` named in message
TC-TRP-08	Scheduling conflict — same driver	Same as above but overlapping driver, different vehicle	Second create fails, 400
TC-TRP-09	Non-overlapping windows allowed	Two trips, same vehicle, back-to-back (no overlap)	Both created successfully
TC-TRP-10	Edit trip while draft	PUT `/trips/:id` while `status: draft`	200, fields updated
TC-TRP-11	Edit trip after dispatch	PUT `/trips/:id` on a `dispatched` trip	400, edit rejected (draft-only rule)
TC-TRP-12	Dispatch happy path	PUT `/trips/:id/dispatch` on a valid draft trip	200, trip → `dispatched`; vehicle & driver → `on_trip`
TC-TRP-13	Dispatch when vehicle no longer available	Vehicle became `in_shop` after trip was drafted, then dispatch attempted	400, BR-3 rejection
TC-TRP-14	Complete happy path	PUT `/trips/:id/complete` with `actualDistance`, `fuelConsumed`, `revenue`	200, trip → `completed`; vehicle odometer incremented; vehicle & driver → `available`
TC-TRP-15	Maintenance-due notification trigger	Complete a trip such that `odometer >= nextMaintenanceMileage - 500`	A `warning` Notification is created
TC-TRP-16	Maintenance-due notification NOT triggered	Complete a trip where resulting odometer is well below threshold	No new notification created
TC-TRP-17	Cancel from draft	PUT `/trips/:id/cancel` on a `draft` trip	200, `status: cancelled`, no assets to release
TC-TRP-18	Cancel from dispatched	Cancel a `dispatched` trip	200, `status: cancelled`; vehicle & driver released to `available`
TC-TRP-19	Cancel a completed trip	Attempt cancel on `completed` trip	400, rejected (terminal state)
TC-TRP-20	Re-transition a cancelled trip	PUT `/trips/:id/status` on a `cancelled` trip	400, terminal-state lock enforced
TC-TRP-21	Audit log written on create	Create a trip, then inspect `AuditLog` collection	Entry with `action: CREATE`, `modelName: Trip`, correct `userId`
TC-TRP-22	Audit log written on dispatch/complete/cancel	Perform each action, inspect audit log	One `UPDATE` entry per action with correct `changes` payload
TC-TRP-23	Trip list filtering	GET `/trips?status=dispatched&priority=high`	Only matching trips returned
TC-TRP-24	Trip code auto-generation	Create a trip without supplying `tripCode`	Response includes a generated `TRIP-XXXXXXXX` code
TC-TRP-25	Pagination	GET `/trips?page=2&limit=10` with >10 trips seeded	Correct page slice, `total` / `page` / `pages` accurate

3.5 Maintenance

ID	Title	Steps	Expected Result
TC-MNT-01	Open maintenance, happy path	POST `/maintenance` on an `available` vehicle	201; vehicle → `in_shop`
TC-MNT-02	BR-9: vehicle on trip	POST `/maintenance` on an `on_trip` vehicle	400, BR-9 message
TC-MNT-03	BR-9: retired vehicle	POST `/maintenance` on a `retired` vehicle	400, BR-9 message
TC-MNT-04	Close maintenance, happy path	PUT `/maintenance/:id/close` on an `open` record	200; vehicle → `available` (unless retired)
TC-MNT-05	Close already-closed record	PUT `/maintenance/:id/close` twice	Second call: 400, “already closed”
TC-MNT-06	Close maintenance on a since-retired vehicle	Vehicle marked `retired` while maintenance was open, then close attempted	200; vehicle status remains `retired` (not forced back to `available`)

3.6 Fuel, Expenses, Reports

ID	Title	Steps	Expected Result
TC-FUEL-01	Create fuel log as financial_analyst	POST `/fuel-expenses/fuel`	201
TC-FUEL-02	Create fuel log as dispatcher	Same body, dispatcher token	403
TC-FUEL-03	Live rates endpoint returns all fuel types	GET `/fuel-expenses/live-rates`	200, keys for diesel/gasoline/cng/electric/hybrid all present, within ±2% of base rates
TC-EXP-01	Create expense with invalid type	`expenseType: "bribe"`	400, Mongoose enum validation error
TC-RPT-01	Dashboard KPIs sum correctly	Seed known trips/costs, GET `/reports/dashboard`	`netProfit` = `totalRevenue` − `totalCosts` exactly
TC-RPT-02	Fuel-efficiency report excludes zero-fuel trips	Trip with `fuelConsumed: 0` completed	Excluded from report (division-by-zero guarded in aggregation `$match`)
TC-RPT-03	Driver-performance access as dispatcher	GET `/reports/driver-performance` with dispatcher token	403
TC-RPT-04	CSV export content-type	GET `/reports/export/trips`	Response is `text/csv`, not JSON; correct column headers

3.7 Notifications

ID	Title	Steps	Expected Result
TC-NOT-01	User sees own + broadcast notifications	Create one user-targeted and one broadcast notification, GET `/notifications` as that user	Both appear
TC-NOT-02	User does not see another user’s targeted notification	Create notification targeted at user B, GET as user A	Not present in user A’s list
TC-NOT-03	Mark as read	PUT `/notifications/:id/read`	200, `read: true` persisted

3.8 Security / Cross-Cutting

ID	Title	Steps	Expected Result
TC-SEC-01	Rate limiting	Send 101 requests to any	101st request → 429

TC-SEC-01	Rate limiting	Send 101 requests to any `/api/*` route within 10 minutes from one IP	101st request → 429
TC-SEC-02	NoSQL injection attempt	Send `{"email": {"$ne": null}, "password": {"$ne": null}}` to login	Rejected (sanitized before reaching query), not authenticated
TC-SEC-03	XSS payload in text field	Create a trip with `notes: "<script>alert(1)</script>"`	Payload sanitized/stored safely, not reflected unescaped
TC-SEC-04	Oversized payload	POST body > 10kb	4xx rejection (`express.json` limit)
TC-SEC-05	JWT tampering	Modify token payload without valid signature	401, “Invalid token”
TC-SEC-06	Server refuses to boot without JWT_SECRET	Start server with `JWT_SECRET` unset	Process exits immediately (verify via startup log/exit code)
TC-SEC-07	`/ai/*` endpoints reachable without auth	Call `/ai/assistant` with no `Authorization` header	200 (confirms the known gap in API Documentation §12-3; treat as a defect to fix, not a pass)

4. Traceability: Business Rules → Test Cases

Rule	Description	Test Case(s)
BR-1	No trip on missing/retired vehicle	TC-TRP-02, TC-TRP-03
BR-2	No trip on missing driver	(covered implicitly by TC-TRP-04/05 setup; add explicit “driver not found” case in automation)
BR-3	Dispatch only if vehicle available	TC-TRP-13
BR-4	No suspended/expired-license driver	TC-TRP-04, TC-TRP-05
BR-5	Cargo ≤ capacity	TC-TRP-06
BR-9	No maintenance on active/retired vehicle	TC-MNT-02, TC-MNT-03
Scheduling conflict	No double-booking	TC-TRP-07, TC-TRP-08, TC-TRP-09
Terminal state lock	No transitions after completed/cancelled	TC-TRP-19, TC-TRP-20
Draft-only edit	Trip immutable once dispatched	TC-TRP-10, TC-TRP-11
Maintenance close guard	No double-close	TC-MNT-05

5. Defect Log Template

ID	Test Case	Severity	Description	Status
DEF-001	TC-DRV-06	Sev-2	Driver `status` enum mismatch between Zod (`on_leave`) and Mongoose (`off_duty`)	Open
DEF-002	TC-SEC-07	Sev-1	`/ai/*` endpoints have no auth middleware	Open

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